

Technical Report: Driving Simulation Telemetry Dashboard & Research Suite

1. Introduction and Project Scope

The **Driving Simulation Telemetry Dashboard** is a specialized, high-performance software application engineered to interface with industry-standard driving simulators such as BeamNG.drive and Assetto Corsa. Its primary objective is to transform raw, high-frequency physics data into actionable intelligence for sim-racers, vehicle dynamics engineers, and behavioral researchers. By acting as a comprehensive data-acquisition (DAQ) system, the application provides a bridge between virtual vehicle physics and professional-grade performance analysis. This report details the application's architecture, its diverse feature set, and the profound analytical value of its exported telemetry datasets.

2. Core Purpose and Strategic Utility

In the realm of competitive sim-racing and automotive research, the margin between success and failure is measured in milliseconds. The strategic purpose of this application is to provide a "second-screen" experience that offers deeper insights than standard in-game HUDs. It serves three distinct use cases:

- * **Performance Optimization:** Allowing drivers to refine their racing lines and pedal inputs through immediate visual feedback.
- * **Engineering Analysis:** Monitoring vehicle health and mechanical stress to optimize setups for longevity and speed.
- * **Human Factor Research:** Utilizing a proprietary simulation engine to study the effects of impairment, fatigue, and aggression on driving safety.

3. Comprehensive Feature Analysis

3.1. Real-Time Telemetry Dashboard

The live dashboard is the application's primary interface, providing a visceral, real-time connection to the virtual vehicle. Beyond standard speed and RPM instrumentation, it features advanced modules designed for technical driving:

- * **The Friction Circle (G-G Diagram):** A critical tool for any performance driver, this module plots Lateral versus Longitudinal G-forces. It allows the driver to visualize the "Traction Circle," ensuring they are maximizing tire grip during the delicate transition from heavy braking to corner entry.
- * **High-Fidelity Input Visualizer:** This module tracks Throttle, Brake, Clutch, and Steering inputs with millisecond precision. It is essential for identifying technical flaws such as "pedal overlap" (simultaneous braking and accelerating) or inconsistent steering rates that can destabilize a vehicle at high speeds.
- * **Live Telemetry Traces:** Scrolling multi-line charts provide immediate historical context for speed and RPM, allowing drivers to see the "shape" of their acceleration and braking curves as they happen.

3.2. Vehicle Health and Tire Management

To support long-distance endurance racing, the application incorporates a sophisticated health monitoring suite. This system tracks the condition of the engine, transmission, suspension, brakes, and aerodynamics on a 0-100% health scale. Additionally, individual tire temperatures and wear levels are monitored for all four corners of the vehicle. This enables drivers to manage heat cycles effectively and predict when a mechanical failure or significant loss of grip is imminent.

3.3. Advanced Behavioral Simulation Engine

A standout feature of the suite is its internal Simulation Mode. Operating independently of any game engine, it utilizes procedural logic to generate realistic telemetry data based on specific "Driver Behavior" profiles. These profiles—including **Professional**, **Reckless**, **Slow**, **Drunk**, and **High**—allow researchers to simulate the impact of various mental and physical states on vehicle control. By introducing artificial "noise" into the steering and delayed reaction times into the pedal logic, the engine provides a safe, reproducible environment for studying impaired driving dynamics.

3.4. Driver Reaction Testing

The application includes a dedicated Reaction Test module that utilizes the physical sim-racing hardware. By measuring the delay between a visual "Stop" stimulus on the screen and the physical activation of the brake pedal, it provides a quantitative metric of driver alertness. This is a vital tool for assessing fatigue levels during multi-hour racing sessions or comparing the natural reflexes of different drivers.

4. Post-Session Analysis and Data Portability

4.1. The Session Analysis Tab

Upon the conclusion of a session, the data is transitioned from live memory to a permanent local database. The Analysis tab provides a browsable history of every drive, accompanied by summary statistics such as total distance, fuel efficiency, and "Coast Time" (the percentage of time spent rolling without pedal input). Interactive, full-session graphs allow for a "post-mortem" of the drive, where specific errors or mechanical failures can be pinpointed on a timeline.

4.2. CSV Data Export and External Integration

For power users, the application offers a robust CSV export function. This feature flattens the complex database structures into a clean, time-indexed spreadsheet compatible with Excel, MATLAB, or Python for advanced statistical modeling. The exported data is divided into several logical categories:

- * **Core Metrics:** Includes `timestamp`, `speed`, `rpm`, and `gear`.
- * **Dynamics:** High-resolution G-force data (`gForceX/Y/Z`) and 3D world coordinates (`x/y/z`).
- * **Health:** Detailed damage coefficients (`damage\_engine`, `damage\_suspension`, etc.) and vital signs like `engineTemp` and `fuel`.

5. Interpreting Advanced Behavioral and Safety Metrics

The true analytical power of the exported dataset lies in the "Derived Metrics" calculated by the application's backend. These variables provide a window into the driver's technique and safety profile:

- * **Smoothness and Stability:** The `jerk\_x` and `jerk\_y` variables measure the rate of change in G-forces. High jerk values indicate erratic inputs, while low values signify the smoothness associated with professional-grade driving.
- * **Vehicle Balance:** Variables such as `oversteer\_correction`, `understeer\_prolong`, and `slip\_angle\_estimate` quantify how the car is behaving relative to the driver's steering input, making it easy to detect when a driver is "pushing" the car beyond its physical limits.
- * **Technical Efficiency:** Metrics like `coasting\_time\_pct` and `pedal\_overlap` help identify inefficient habits. High coasting time can indicate a lack of aggression or an attempt at fuel saving, while high pedal overlap often points to a lack of technical discipline during downshifts.

6. Conclusion

The Driving Simulation Telemetry Dashboard is more than a visual aid; it is a comprehensive diagnostic platform. By combining real-time visualization with deep-dive analytical exports and behavioral simulation, it provides a holistic view of the driving experience. Whether used to shave tenths of a second off a lap time or to conduct serious research into driver impairment, the application delivers the precision and depth required for professional telemetry analysis in the modern era.

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